

Selected Research in Spatial Intelligence & Real-Time Perception

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Researcher and engineer focused on scalable real-time perception, probabilistic tracking, spatial reasoning, and real-time AI systems for physical environments. My work spans GPU-accelerated multi-object tracking, robotics perception, world-model-based multimodal learning, and efficient large-scale training systems.

1 GPU-Accelerated Multi-Sensor Perception and Multi-Object Tracking

Developed a GPU-accelerated multi-object tracking system for real-time geospatial perception across distributed camera networks (refer figure 2).

The system integrates probabilistic 2D-to-3D localization, uncertainty-aware geospatial detection, and GPU-accelerated multi-hypothesis tracking to infer world-space trajectories under clutter, occlusion, and partial observability (refer figure 1).

Key contributions:

- Developed a **scalable spatial perception system** that **converts raw distributed image-space detections into uncertainty-aware world-space object trajectories** through probabilistic geometric reasoning and multi-sensor fusion [1].
- Designed a **parallelized GLMB-based tracking framework** implemented in PyTorch for scalable hypothesis generation and data association.
- Engineered a **probabilistic camera projection pipeline** converting image-space detections into uncertainty-aware global coordinates [2].
- Modeled **spatial uncertainty, occlusion, and clutter** to improve robustness in real-world multi-view perception scenarios.
- Achieved **sub-100 ms tracking latency** across Jetson Orin, RTX 4090, and L40S GPU environments (refer figure 3) [3].
- Evaluated scalability across target cardinality, hypothesis growth, clutter intensity, and detection noise using OSPA-based tracking metrics.

2 Autonomous Human Following Robot

Built an end-to-end ROS-based autonomous perception and control system for real-time human following on embedded robotic hardware.

System capabilities:

- Integrated YOLOv3-tiny detection, SORT-based tracking, and custom appearance-based re-identification.
- Developed **modular ROS perception-to-control** architecture with autonomous angular and lateral control.
- Deployed on **Jetson-based robotic hardware** with real-time 10–15 FPS perception and closed-loop navigation.

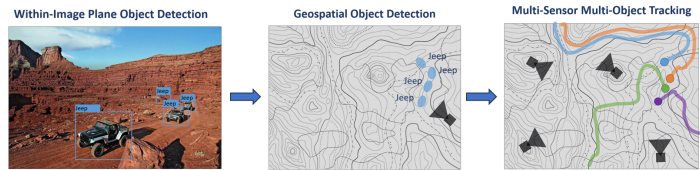


Figure 1: We developed a **multi-sensor spatial perception pipeline** that transforms **image-space detections** from distributed visual sensors into **uncertainty-aware world-space object trajectories**. The system combines deep-learning-based object detection, probabilistic camera projection, geospatial localization, multi-camera sensor fusion, and GPU-accelerated multi-hypothesis tracking to enable scalable real-time perception in complex physical environments.

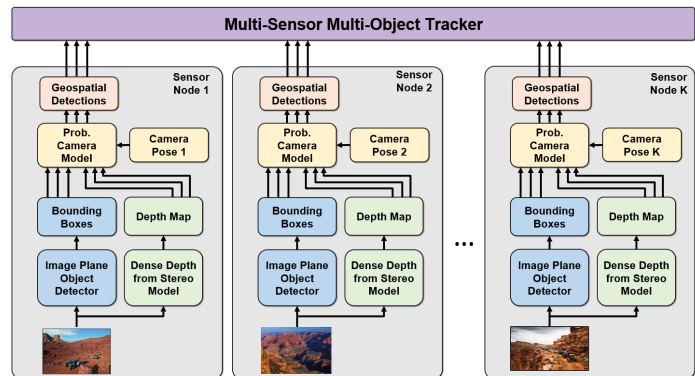


Figure 2: **Multi-sensor spatial perception pipeline** integrating deep-learning-based object detection, probabilistic geospatial localization, sensor fusion, and GPU-accelerated multi-object tracking.

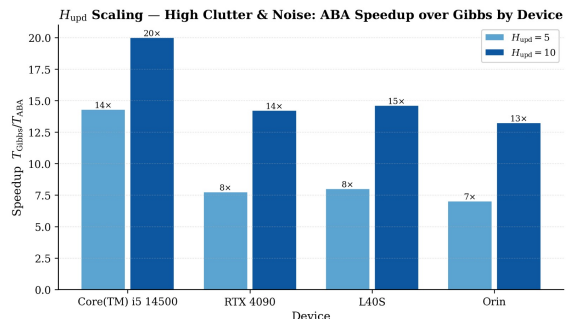


Figure 3: Preliminary results: Our custom parallelized association formulation (ABA) achieves substantial acceleration over standard sequential Gibbs-based GLMB inference across edge, desktop, and server-class GPU platforms. Here, (H_{upd}) denotes the number of association hypotheses generated per timestep, directly controlling tracker complexity.

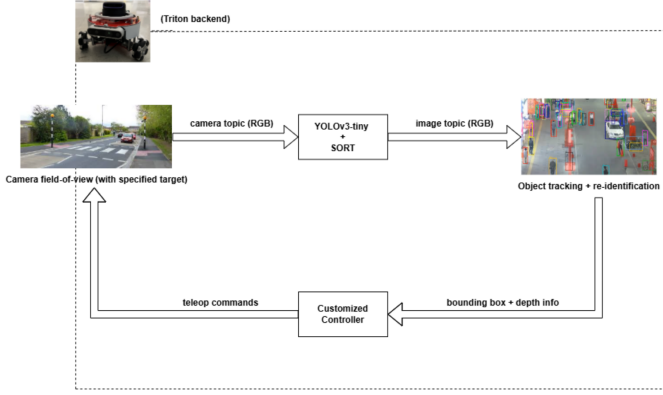
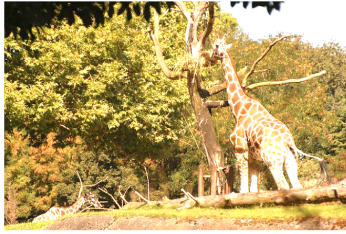


Figure 4: End-to-end perception-tracking-control pipeline under ROS Noetic



Q: Is there an animal in the right half of this image?
A: Yes, there is an animal in the right half of this image. It appears to be a giraffe standing near a tree.

Figure 5: Evaluated whether self-supervised world models produce stronger spatial awareness and geometric scene understanding than conventional vision-language encoders.

3 World-Model-Based Spatial Reasoning in VLMs

Investigated whether self-supervised world models provide **stronger spatial reasoning** capabilities than CLIP-style semantic encoders in multimodal systems.

Core idea: Replace conventional vision-language alignment encoders with I-JEPA representations trained to model geometric and structural continuity in visual scenes.

Key contributions:

- Designed a **multimodal architecture** integrating frozen I-JEPA visual representations with Qwen2 language models.
- Developed a **feature-alignment pipeline** bridging geometric latent representations and language embedding spaces.
- Implemented efficient LoRA and quantized training pipelines enabling multimodal experimentation on limited hardware.
- **Evaluated spatial grounding** and hallucination behavior using POPE-based object permanence benchmarks.

4 Efficient Transformer Scaling (DeepMind)

Investigated efficient transformer scaling strategies for increasing representational capacity without proportional compute growth.

- Built a **unified PyTorch-based pretraining pipeline** for systematic evaluation of **residual-stream scaling** architectures (mHC, VWN, AltUp), training 135M-parameter language models on 13.5B tokens under controlled experimental settings.
- Developed an evaluation framework benchmarking model performance across **training/inference FLOPs, runtime overhead, throughput, and downstream language understanding tasks**.
- Identified major bottlenecks where architectures with minimal additional compute requirements produced up to **300% longer training times**, tracing inefficiencies to poor memory utilization and CUDA kernel launch overhead.
- Demonstrated up to **30.4% improvement on downstream language benchmarks** while maintaining near-constant parameter and FLOP growth, revealing practical pathways for efficient transformer scaling (refer Table 1).

Architecture	Train Hrs. (H100)	Train Loss	Val bpb	#Params	FLOPS/token	CORE Score
Nanochat (cleaned)	7	2.658	0.810369	135,266,304	7.60E+08	0.1644
AltUp	14.5	2.635	0.799419	135,266,376	7.60E+08	0.1764
VWN	18.5	2.616	0.799419	137,713,153	7.74E+08	0.1820
mHC	19.6	2.626	0.820319	137,036,424	7.59E+08	0.2144
2x-Naive	21	2.460	0.750005	440,401,920	2.54E+09	0.2268

Table 1: Controlled evaluation of residual-stream scaling architectures across training efficiency, FLOPs, throughput, and downstream language understanding performance.

Publications

- [1] P. Balakrishnan, S. Barik, S. M. O’Rourke, and B. M. Marlin, “Gpu-glmb: Assessing the scalability of gpu-accelerated multi-hypothesis tracking,” in *IEEE Military Communications Conference (MILCOM)*, 2025.
- [2] B. M. Marlin, S. M. O’Rourke, S. Barik, P. Balakrishnan, and C. Samplawski, “Improving target tracking in stone soup via state-dependent process noise models,” Under review at International Conference on Information Fusion, 2026, 2026.
- [3] S. Barik, P. Balakrishnan, S. M. O’Rourke, and B. M. Marlin, “Gpu-accelerated generalized labeled multi-bernoulli filter for multi-detection point targets,” Under review at IEEE Transactions on Signal Processing.

Technical Stack

- **Spatial Perception & Geometry:** Multi-view geometry, probabilistic camera models, geospatial localization, sensor fusion, uncertainty-aware tracking, spatial grounding
- **Programming Languages & Frameworks:** Python, SQL, C++, R, Hive, PyTorch, NumPy, Pandas, PySpark, ROS